
Uso de Ambari para Monitorización y Administración

Agenda

- ¿Qué es Ambari?
- Administrando Hadoop
- Alertas, Servicios y Hosts
- Vistas de Usuario

¿Qué es Ambari?

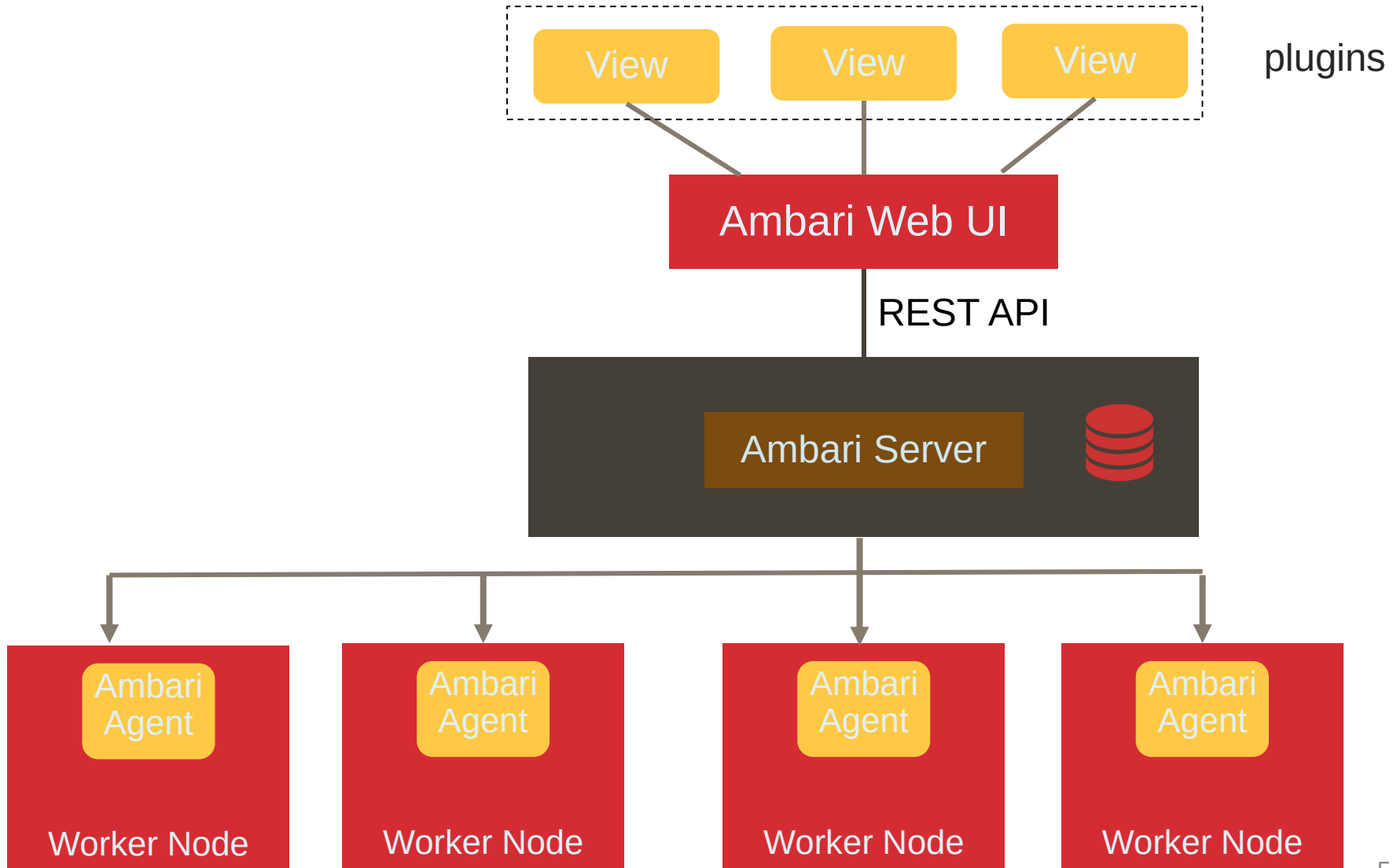
- Framework 100% Software Libre para aprovisionar, administrar y monitorizar clusters Apache Hadoop
- Aprovisionamiento
 - Proporciona asistentes paso a paso para instalar servicios Hadoop a través de hosts
 - Maneja la configuración de servicios
- Administración
 - Proporciona un punto de administración central para iniciar, detener y reconfigurar servicios a través de todo el cluster
- Monitorización
 - Proporciona cuadros de mando de monitorización de estado
 - Ambari Metrics System
 - Ambari Alert Framework

Características



- ❖ Instalación por asistentes
- ❖ Instalación de cluster a través de API no interactiva
- ❖ Control Granular de servicios
- ❖ Configuración de Servicio de Cluster
- ❖ Monitorización y Alertas
- ❖ REST API para integración
- ❖ Ambari Views para plug-ins

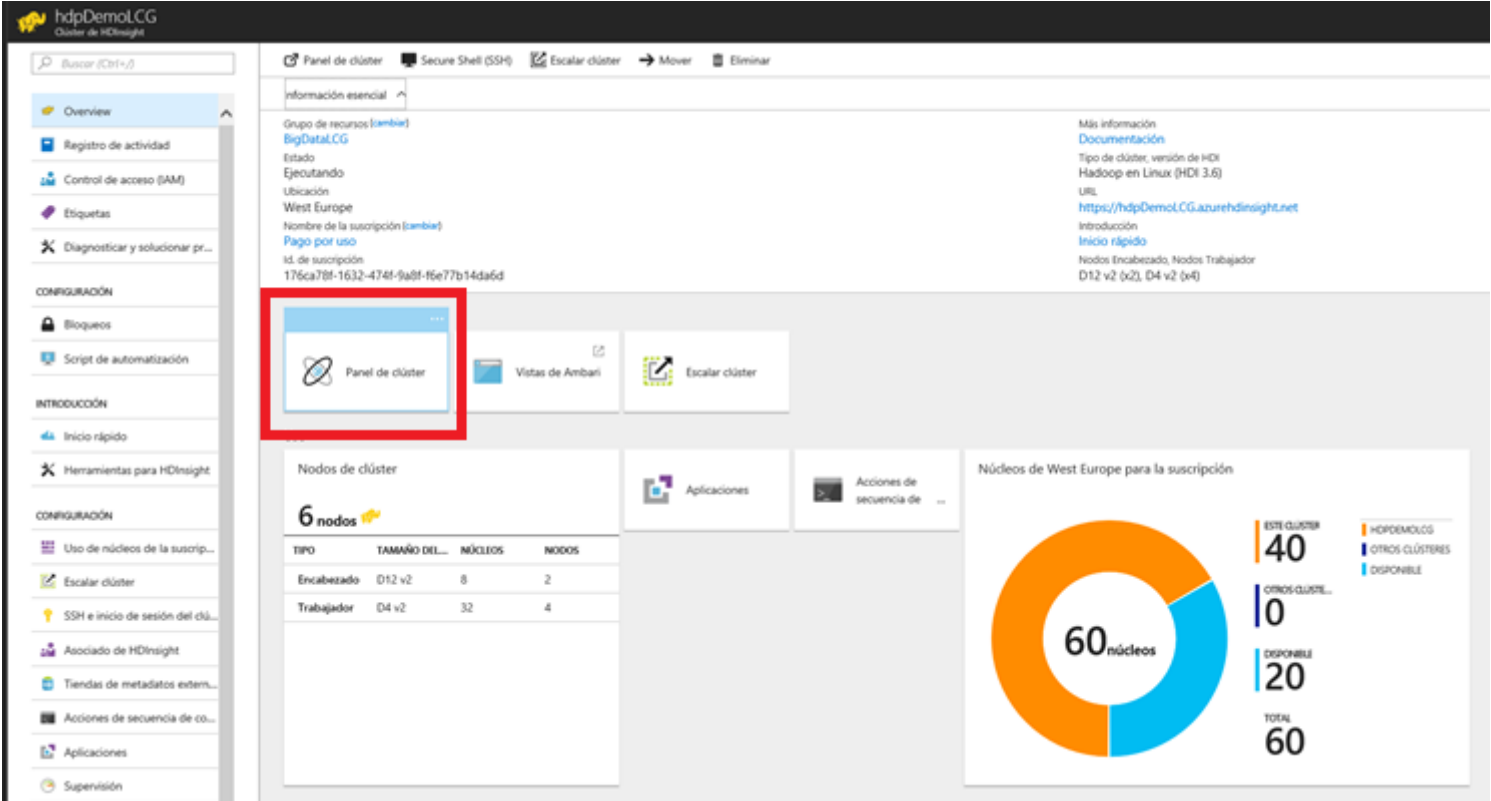
Arquitectura



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El interfaz



hdpDemo1.CG
Clúster de HDInsight

Buscar (Ctrl+F)

Panel de clúster Secure Shell (SSH) Escalar clúster Mover Eliminar

información esencial

Grupo de recursos [\[cambiar\]](#)
BigData1.CG
Estado: Ejecutando
Ubicación: West Europe
Nombre de la suscripción [\[cambiar\]](#)
[Pago por uso](#)
Id. de suscripción: 176ca78f-1632-474f-9a8f-f6e77b14da6d

Más información
[Documentación](#)
Tipo de clúster, versión de HDI
Hadoop en Linux (HDI 3.6)
URL: <https://hdpDemo1.CG.azurehdinsight.net>
[Introducción](#)
[Inicio rápido](#)
Nodos Encabezado, Nodos Trabajador
D12 v2 (x2), D4 v2 (x4)

Panel de clúster Vistas de Ambari Escalar clúster

Nodos de clúster
6 nodos

TIPO	TAMAÑO DEL...	NÚCLEOS	NODOS
Encabezado	D12 v2	8	2
Trabajador	D4 v2	32	4

Aplicaciones Acciones de secuencia de ...

Núcleos de West Europe para la suscripción

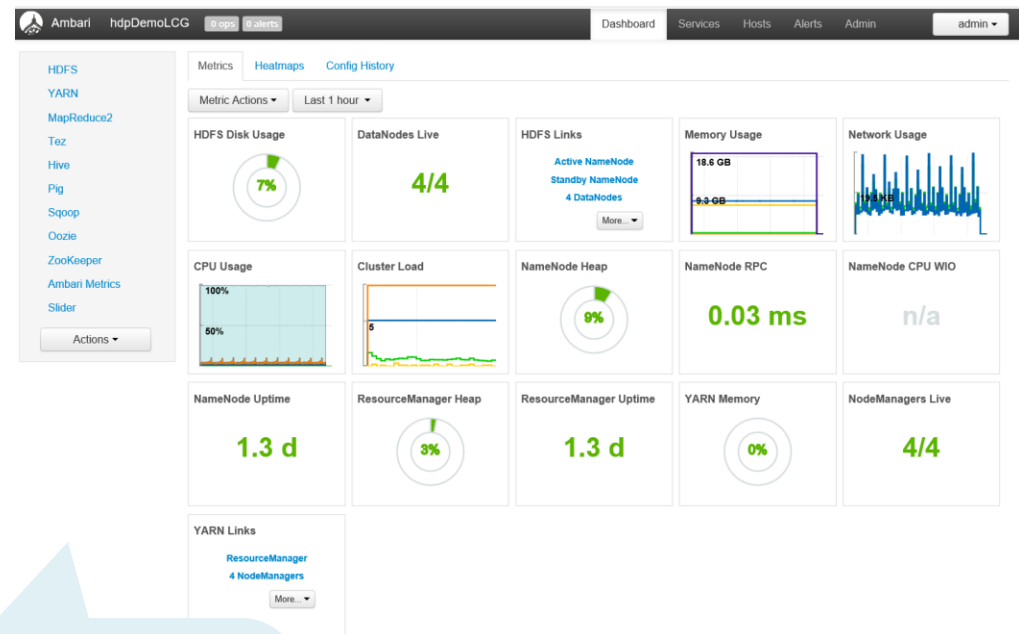
ESTE CLÚSTER: 40
OTROS CLÚSTER...: 0
DISPONIBLE: 20
TOTAL: 60

60 núcleos

Acceso desde el portal Azure

Cuadro de Mando de métricas

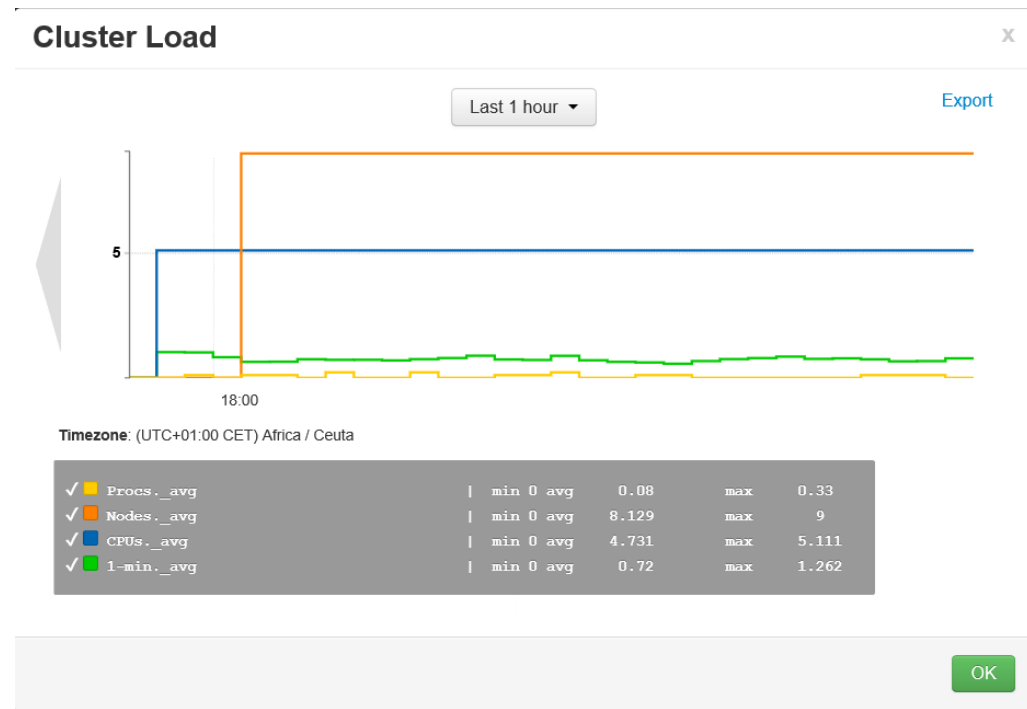
- La pestaña de métricas del Cuadro de Mando muestra métricas a nivel de cluster:
 - Uso CPU
 - Uso disco HDFS
 - Uso Memoria
 - Uso Red
 -
- Nos permite conocer el estado de un vistazo
- Puede personalizarse agregando o quitando widgets



- La lista de servicios instalados siempre aparece en el panel de la izquierda.
- El botón de Acciones permite iniciar, detener y reiniciar el cluster

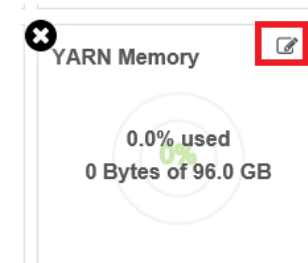
Detalle

- Podemos obtener más detalle de:
 - Uso CPU
 - Carga del Cluster
 - Uso Red
 - Uso de Memoria
- Las estadísticas de uso pueden verse por periodos de tiempo
- Para otras métricas podemos ver información adicional pasando el ratón



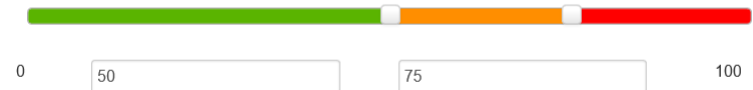
Personalización

- Podemos personalizar
 - YARN Memory
 - Node Managers
 - Resource Managers
 - NameNode CPU
 - NameNode RPC
 - NameNode Heap



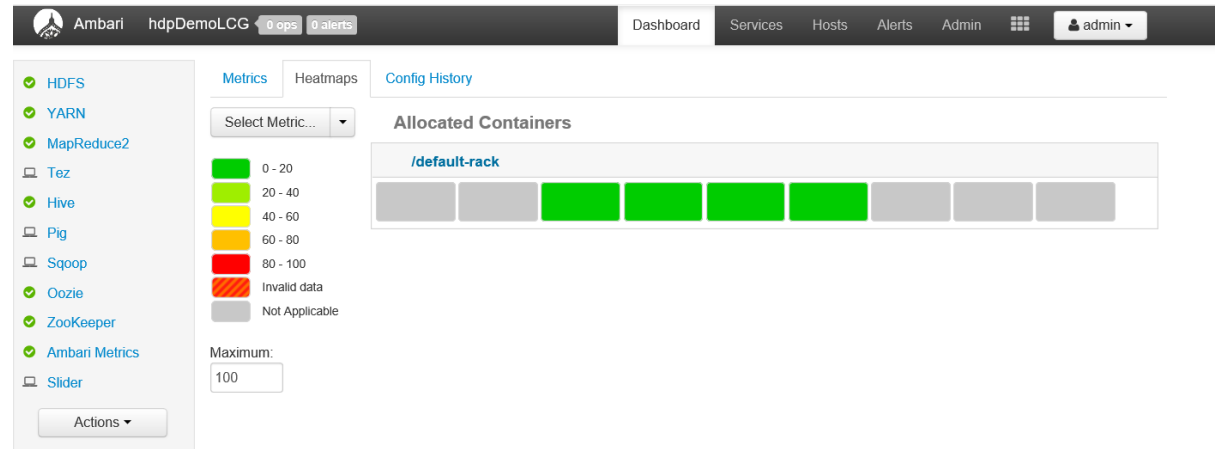
Customize Widget

Edit the percentage thresholds to change the color of current pie chart.
Enter two numbers between 0 to 100



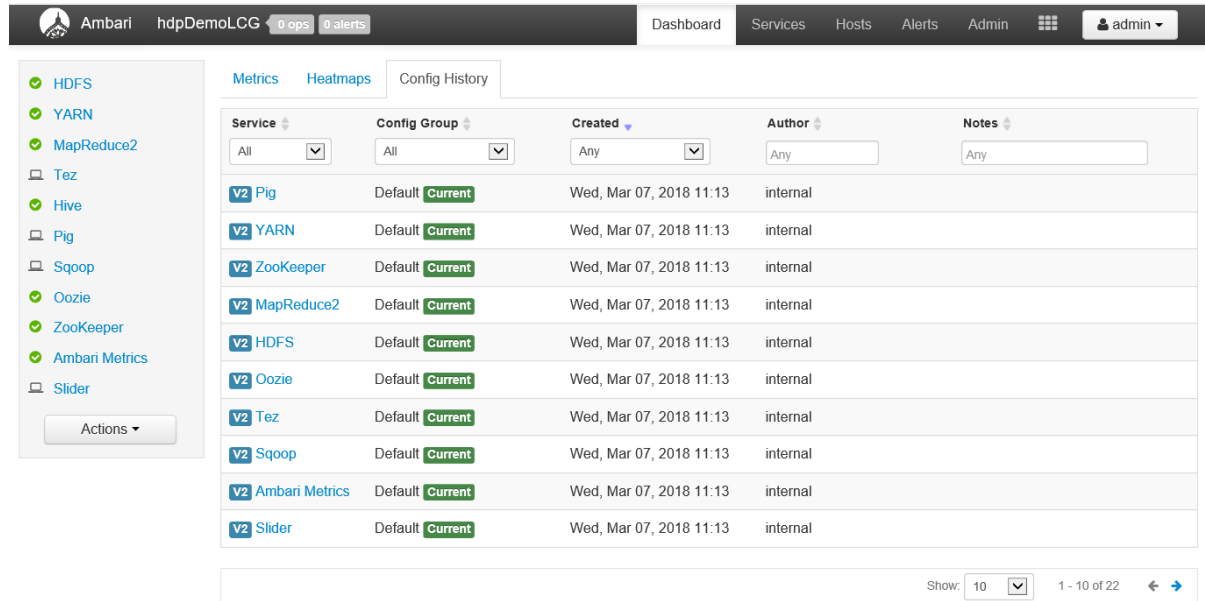
HeatMap

- Vista en color de cada nodo del cluster
- Podemos seleccionar métricas



Historia de Configuración

- Lista del histórico de cambios
- Podemos bajar al detalle de cada servicio



The screenshot shows the Ambari web interface. The top navigation bar includes the Ambari logo, the instance name 'hdpDemoLCG', and status indicators for '0 ops' and '0 alerts'. The main navigation tabs are 'Dashboard', 'Services', 'Hosts', 'Alerts', and 'Admin'. The user is logged in as 'admin'. On the left sidebar, a list of services is shown with green checkmarks: HDFS, YARN, MapReduce2, Tez, Hive, Pig, Sqoop, Oozie, ZooKeeper, Ambari Metrics, and Slider. An 'Actions' button is at the bottom of the sidebar. The main content area is titled 'Config History' and contains a table with the following columns: Service, Config Group, Created, Author, and Notes. The table lists 11 services, all with a 'Default' config group and a 'Current' status. The 'Created' column shows the date 'Wed, Mar 07, 2018 11:13' for all entries, and the 'Author' column shows 'internal'. At the bottom right, there is a 'Show: 10' dropdown and a pagination indicator '1 - 10 of 22'.

Service	Config Group	Created	Author	Notes
V2 Pig	Default	Current	Wed, Mar 07, 2018 11:13	internal
V2 YARN	Default	Current	Wed, Mar 07, 2018 11:13	internal
V2 ZooKeeper	Default	Current	Wed, Mar 07, 2018 11:13	internal
V2 MapReduce2	Default	Current	Wed, Mar 07, 2018 11:13	internal
V2 HDFS	Default	Current	Wed, Mar 07, 2018 11:13	internal
V2 Oozie	Default	Current	Wed, Mar 07, 2018 11:13	internal
V2 Tez	Default	Current	Wed, Mar 07, 2018 11:13	internal
V2 Sqoop	Default	Current	Wed, Mar 07, 2018 11:13	internal
V2 Ambari Metrics	Default	Current	Wed, Mar 07, 2018 11:13	internal
V2 Slider	Default	Current	Wed, Mar 07, 2018 11:13	internal

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Alertas

- Nos muestra alertas en la parte superior de la página

The screenshot shows the Ambari Alerts page for 'Host Disk Usage'. The top navigation bar includes 'Dashboard', 'Services', 'Hosts', 'Alerts', and 'Admin'. A red badge indicates '1 alert'. The main content area is titled 'Host Disk Usage' with a green 'OK (24)' status badge. Below the title is a 'Configuration' section with a description: 'This host-level alert is triggered if the amount of disk space used goes above specific thresholds. The default threshold values are 50% for WARNING and 80% for CRITICAL.' The 'Check Interval' is set to '1 Minute'. To the right, the alert details are listed: 'State: Enabled', 'Service: Ambari', 'Component: Ambari Agent', 'Type: SCRIPT', 'Groups: AMBARI Default', 'Last: Mon, Apr 04, 2016', and 'Changed: 21:49'. Below the configuration is an 'Instances' table with columns: 'Service / Host', 'Status', '24-Hour', and 'Response'.

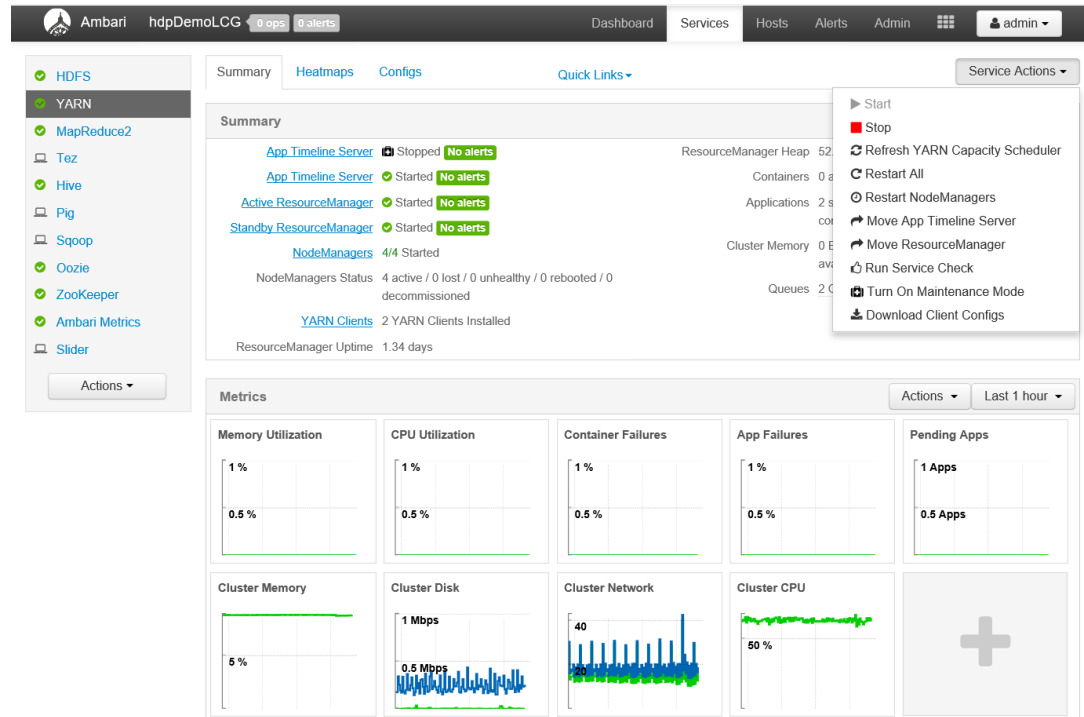
Service / Host	Status	24-Hour	Response
Ambari / hn0-ntlive.kqrgngcrnh0eria3ub12ko3iad.bx.internal.cloudapp.net	OK for 9 hours	1	Capacity Used: [1.07%, 11.3 GB], Capacity Total: [1.1 T...
Ambari / hn1-ntlive.kqrgngcrnh0eria3ub12ko3iad.bx.internal.cloudapp.net	OK for 9 hours	1	Capacity Used: [1.11%, 11.8 GB], Capacity Total: [1.1 T...
Ambari / wn0-ntlive.kqrgngcrnh0eria3ub12ko3iad.bx.internal.cloudapp.net	OK for 2 hours	2	Capacity Used: [1.39%, 14.7 GB], Capacity Total: [1.1 ...

The screenshot shows the Ambari Alerts page with a modal window titled '1 Critical or Warning Alerts'. The modal displays a table with columns: 'Service / Host', 'Alert Definition Name', and 'Status'. The first row shows 'Ambari' for 'Ambari Server Alerts' with a 'CRIT' status for 2 hours. Below the table, it says 'There are 15 stale alerts from 15 ho...'. The modal also includes a 'Show' dropdown set to '10', a '1 - 1 of 1' indicator, and a 'Go to Alerts Definitions' button. The background shows a list of alert definitions with columns for 'Alert Definition Name', 'Status', and 'Last'.

Service / Host	Alert Definition Name	Status
Ambari	Ambari Server Alerts	CRIT for 2 hours

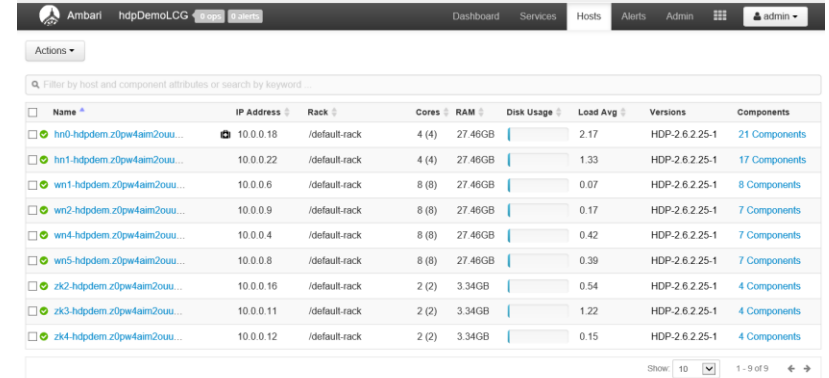
Servicios

- La página de servicios nos proporciona información sobre el estado de los servicios en ejecución
- Los iconos indican el estado o acciones que debemos de tomar
- Para cada servicio tenemos una lista de acciones



Hosts

- Métricas a nivel de sistema para cada nodo
- Podemos ver la lista de componentes
- Detalle del host



Ambari hdpDemoLCG

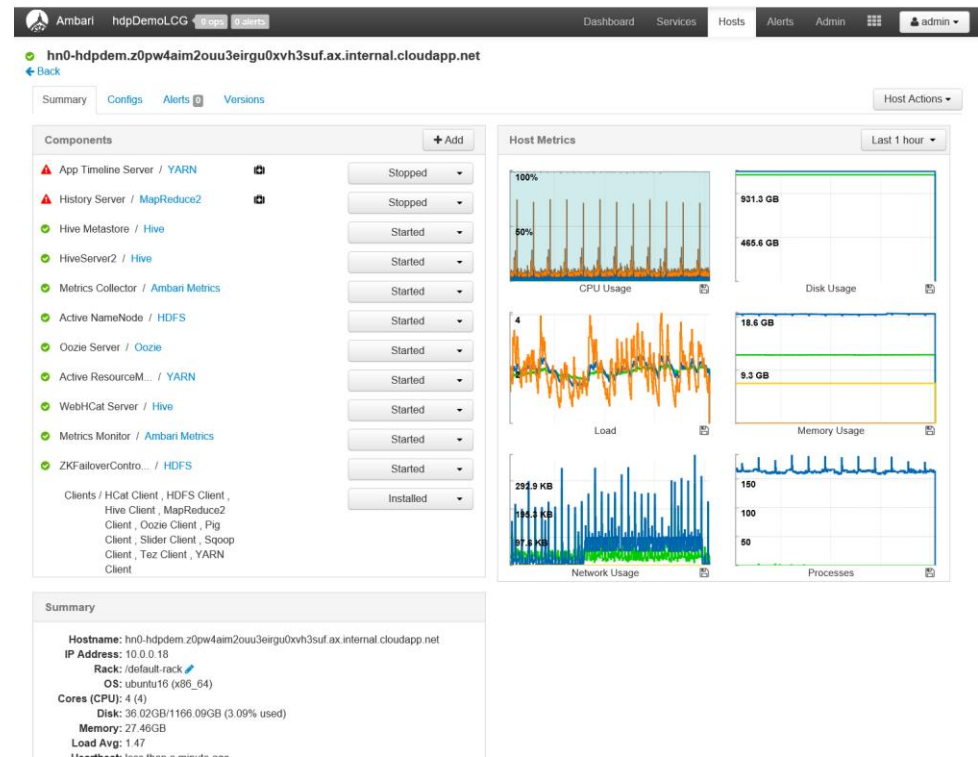
Dashboard Services Hosts Alerts Admin

Actions

Filter by host and component attributes or search by keyword.

Name	IP Address	Rack	Cores	RAM	Disk Usage	Load Avg	Versions	Components
hn0-hdpdem.z0pw4aim2ou...	10.0.0.18	/default-rack	4 (4)	27.46GB		2.17	HDP-2.6.2.25-1	21 Components
hn1-hdpdem.z0pw4aim2ou...	10.0.0.22	/default-rack	4 (4)	27.46GB		1.33	HDP-2.6.2.25-1	17 Components
wn1-hdpdem.z0pw4aim2ou...	10.0.0.6	/default-rack	8 (8)	27.46GB		0.07	HDP-2.6.2.25-1	8 Components
wn2-hdpdem.z0pw4aim2ou...	10.0.0.9	/default-rack	8 (8)	27.46GB		0.17	HDP-2.6.2.25-1	7 Components
wn4-hdpdem.z0pw4aim2ou...	10.0.0.4	/default-rack	8 (8)	27.46GB		0.42	HDP-2.6.2.25-1	7 Components
wn5-hdpdem.z0pw4aim2ou...	10.0.0.8	/default-rack	8 (8)	27.46GB		0.39	HDP-2.6.2.25-1	7 Components
zk2-hdpdem.z0pw4aim2ou...	10.0.0.16	/default-rack	2 (2)	3.34GB		0.54	HDP-2.6.2.25-1	4 Components
zk3-hdpdem.z0pw4aim2ou...	10.0.0.11	/default-rack	2 (2)	3.34GB		1.22	HDP-2.6.2.25-1	4 Components
zk4-hdpdem.z0pw4aim2ou...	10.0.0.12	/default-rack	2 (2)	3.34GB		0.15	HDP-2.6.2.25-1	4 Components

Show 10 1 - 9 of 9



Ambari hdpDemoLCG

Dashboard Services Hosts Alerts Admin

hn0-hdpdem.z0pw4aim2ou3eiru0xvh3suf.ax.internal.cloudapp.net

Back

Summary Configs Alerts Versions Host Actions

Components

- App Timeline Server / YARN Stopped
- History Server / MapReduce2 Stopped
- Hive Metastore / Hive Started
- HiveServer2 / Hive Started
- Metrics Collector / Ambari Metrics Started
- Active NameNode / HDFS Started
- Oozie Server / Oozie Started
- Active ResourceM... / YARN Started
- WebHCat Server / Hive Started
- Metrics Monitor / Ambari Metrics Started
- ZKFailoverContro... / HDFS Started
- Clients / HCat Client, HDFS Client, Hive Client, MapReduce2 Client, Oozie Client, Pig Client, Slider Client, Sqoop Client, Tez Client, YARN Client Installed

Host Metrics

CPU Usage: 100% (graph)

Disk Usage: 931.3 GB (graph)

Load: 4 (graph)

Memory Usage: 18.6 GB (graph)

Network Usage: 294.9 KB (graph)

Processes: 150 (graph)

Summary

Hostname: hn0-hdpdem.z0pw4aim2ou3eiru0xvh3suf.ax.internal.cloudapp.net

IP Address: 10.0.0.18

Rack: /default-rack

OS: ubuntu16 (x86_64)

Cores (CPU): 4 (4)

Disk: 36.02GB/1166.09GB (3.09% used)

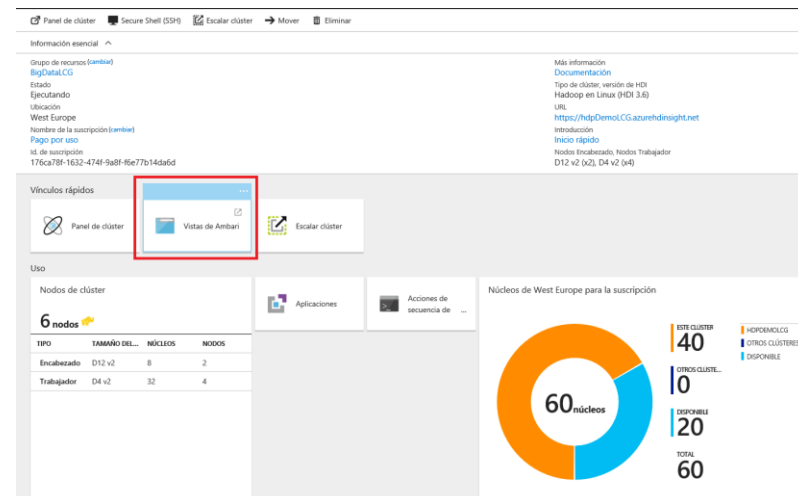
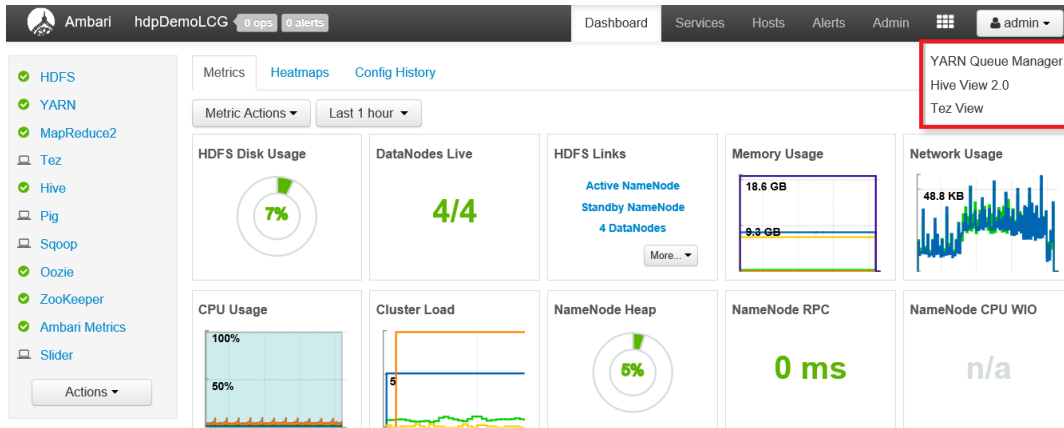
Memory: 27.46GB

Load Avg: 1.47

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Vistas de Usuario



Vistas de Usuario: YARN

The screenshot displays the Ambari web interface for YARN configuration. The top navigation bar includes the Ambari logo, the name 'hdpDemoLGP', and status indicators for '0 ops' and '0 alerts'. The main menu contains links for 'Dashboard', 'Services', 'Hosts', 'Alerts', and 'Admin', with the 'admin' user selected.

The left sidebar contains a '+ Add Queue' button and an 'Actions' dropdown. Below this, a list of queues is shown: 'root (100%)' (100% capacity), 'default (95%)' (95% capacity), and 'joblauncher (5%)' (5% capacity). The 'Scheduler' section is expanded, showing configuration for the 'root' queue: 'Maximum Applications' (10000), 'Maximum AM Resource' (33%), 'Node Locality Delay' (0), 'Calculator' (Default Resource Calculator), 'Queue Mappings' (empty), and 'Queue Mappings Override' (Disabled). The 'Versions' section at the bottom left shows two versions: 'v2' (Current, TOPOLOGY_RESOLVED) and 'v1' (INITIAL).

The main content area is titled 'root' and shows the 'Capacity' section with a 'Level Total' bar at 100%. Below this, the 'root' queue's 'Capacity' is set to 100%, and the 'Max Capacity' is also 100%. The 'Access Control and Status' section shows the 'State' as 'Running' and 'Stopped'. The 'Administer Queue' section has 'Effective Administrators' set to 'Anyone'. The 'Submit Applications' section has 'Effective Users' set to 'Anyone'. The 'Resources' section on the right shows 'User Limit Factor' (1), 'Minimum User Limit' (100%), 'Maximum Applications' (Inherited), 'Maximum AM Resource' (Inherited), and 'Priority' (0).

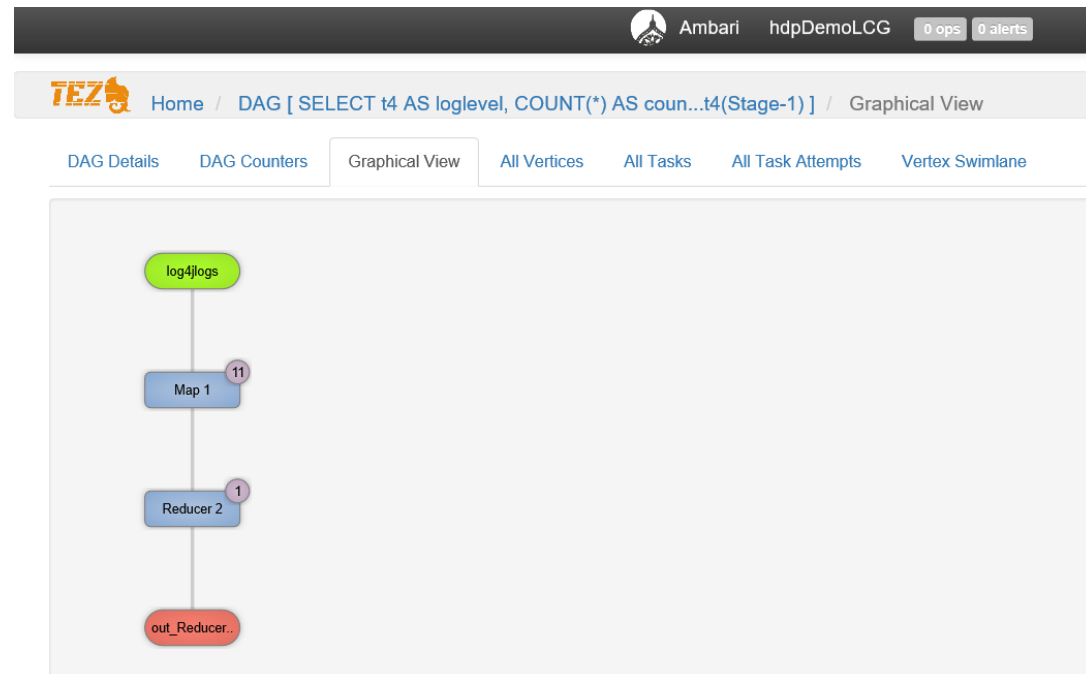
- Uso de las colas y configuración
- Podemos definir nuevas colas

Vistas de Usuario: TEZ

The screenshot shows the Ambari TEZ interface. At the top, there's a navigation bar with 'Ambari', 'hdpDemoLCG', and '0 ops 0 alerts'. Below it, the 'TEZ' logo and 'Home / All Queries' are visible. A search bar is present with fields for Query ID, User, DAG ID, Tables Read, Tables Written, App ID, Queue, and Execution Mode. A table below lists query details:

Query ID	User	Status	Query	DAG ID	Tables Read	Tables Written	LLAP App ID	Start Time	End Time	Duration	Application Id	Queue
hive_20180307103731_4d52594a-8...	admin	SUCCEEDED	SELECT t4 AS logl...	dag_1520417650442_0002_1	default.log4logs	None	Not Available	07 Mar 2018 11:37:31	07 Mar 2018 11:37:58	26s 477ms	application_152041...	default

- Muestra todos los DAG en un periodo de tiempo
- Podemos acceder al detalle



Vistas de Usuario: Hive

Ambari

hdpDemoLCG

0 ops

0 alerts

Dashboard

Services

Hosts

Alerts

Admin

admin

HIVE

+ NEW JOB

+ NEW TABLE

NOTIFICATIONS

QUERY

JOBS

TABLES

SAVED QUERIES

UDFs

SETTINGS

Worksheet1 *

+

DATABASE

Select or search database/schema

default

Browse

1 select * from hivesampletable;

Execute

Save As

Insert UDF

Visual Explain

RESULTS

LOG

VISUAL EXPLAIN

TEZ UI

Filter columns

×

≡

←

→

↗

hivesampletable.clientid	hivesampletable.querytime	hivesampletable.market	hivesampletable.deviceplatform	hivesampletable.devicemake	hivesampletable.devicemodel	hivesampletable.state	hivesampletable.country	hivesampletable
8	18:54:20	en-US	Android	Samsung	SCH-I500	California	United States	13.9204007
23	19:19:44	en-US	Android	HTC	Incredible	Pennsylvania	United States	null

default

Tables(1)

Search Tables

hivesampletable



Navegando por interfaz Ambari

